Lecture 7 recap

In Lecture 7:

- Revised power series, including Taylor series representations of functions.
- Saw an example of how to solve an ODE using a power series.
- Revised the ratio test to determine the radius of convergence of a series.

Lecture 8

In Lecture 8, we will:

1. Learn the power series solution methodology
2. Solve the Airy differential equation

Power series solution methodology

1. Substitute $y(t) = \sum_{n=0}^{\infty} c_n t^n$ into ODE.
2. Change index in each sum to have the same power of t.
3. Extract any low index terms so all sums are over the same range.
4. Solve for the coefficients using recurrence relations.
5. Check if you have the right number of linearly independent solutions.

Exercise 2.6: Find series solutions of $y''(t) + y(t) = 0$

We know two linearly independent solutions are $\cos(t)$ and $\sin(t)$.

Substituting $y(t) = \sum_{n=0}^{\infty} c_n t^n$ into the ODE gives

$$\sum_{n=2}^{\infty} c_n n(n-1)t^{n-2} + \sum_{n=0}^{\infty} c_n t^n = 0$$

In the first sum let $m = n - 2$ and in the second let $m = n$, then

$$\sum_{m=-2}^{\infty} \underbrace{c_{m+2}(m+2)(m+1)t^m}_{=0 \text{ for } m=-2,-1} + \sum_{m=0}^{\infty} c_m t^m = 0$$

$$\sum_{m=0}^{\infty} \left\{ c_{m+2}(m+2)(m+1) + c_m \right\} t^m = 0$$

We require

$$c_{m+2}(m+2)(m+1) + c_m = 0 \quad \Rightarrow \quad c_{m+2} = -\frac{c_m}{(m+2)(m+1)} \quad \text{for} \quad m = 0, 1, 2, \ldots$$

a recurrence relation.

$$c_2 = -\frac{c_0}{2}\,, \quad c_4 = -\frac{c_2}{4 \cdot 3} = \frac{c_0}{4 \cdot 3 \cdot 2}\,, \ldots, \quad c_{2n} = \frac{(-1)^n c_0}{(2n)!}$$

$$c_3 = -\frac{c_1}{3 \cdot 2}\,, \quad c_5 = -\frac{c_3}{5 \cdot 4} = \frac{c_1}{5 \cdot 4 \cdot 3 \cdot 2}\,, \ldots, \quad c_{2n+1} = \frac{(-1)^n c_1}{(2n+1)!}$$

We have found two linearly independent series solutions (with $R = \infty$)

$$c_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} t^{2n} = c_0 \underbrace{\left(1 - \frac{t^2}{2!} + \dots\right)}_{= \cos(t)}$$

$$\text{and} \quad c_1 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} t^{2n+1} = c_1 \underbrace{\left(t - \frac{t^3}{3!} + \dots\right)}_{= \sin(t)}$$

Replace c_0 and c_1 with C_1 and C_2, then the general solution is

$$\begin{aligned} y(t) &= C_1 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} t^{2n} + C_2 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} t^{2n+1} \\ &= C_1 \cos(t) + C_2 \sin(t) \end{aligned}$$

A more challenging problem

Airy differential equation

- Is the 2nd-order, homogeneous linear ODE

$$y''(x) - x\,y(x) = 0$$

- It arises when studying optical phenomena, among other things.
- No non-trivial solutions in terms of elementary functions.

Search for a power series solution: $y(x) = \sum_{n=0}^{\infty} c_n x^n$

$$\sum_{n=0}^{\infty} c_n n(n-1) x^{n-2} - \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

Use $n - 2 \to m + 1$ in first sum

$$\Rightarrow \quad \sum_{m=-3}^{\infty} c_{m+3}(m+3)(m+2)x^{m+1} - \sum_{m=0}^{\infty} c_m x^{m+1} = 0$$

$$\Rightarrow \quad 2\,c_2 + \sum_{m=0}^{\infty} \{c_{m+3}(m+3)(m+2) - c_m\}\, x^{m+1} = 0$$

$$\Rightarrow \quad c_2 = 0 \qquad \text{and} \qquad c_{m+3} = \frac{c_m}{(m+3)(m+2)}$$

$$c_3 = \frac{1}{2 \cdot 3}\, c_0 \;, \ldots, \quad c_{3k} = \frac{1}{2 \cdot 3 \cdot 5 \cdot 6 \cdots (3k-1) \cdot 3k}\, c_0$$

$$c_4 = \frac{1}{3 \cdot 4}\, c_1 \;, \ldots, \quad c_{3k+1} = \frac{1}{3 \cdot 4 \cdot 6 \cdot 7 \cdots 3k \cdot (3k+1)}\, c_1$$

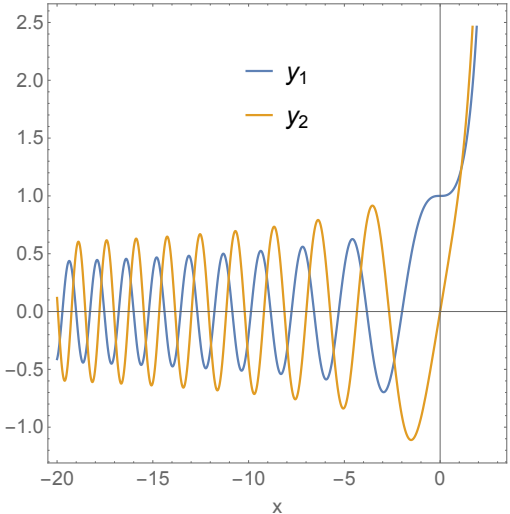
Two linearly independent solutions (with $R = \infty$):

$$y_1(x) = 1 + \sum_{k=1}^{\infty} \frac{x^{3k}}{2 \cdot 3 \cdot 5 \cdot 6 \cdots (3k-1) \cdot 3k}$$

$$y_2(x) = x + \sum_{k=1}^{\infty} \frac{x^{3k+1}}{3 \cdot 4 \cdot 6 \cdot 7 \cdots 3k \cdot (3k+1)}$$

The general solution of Airy's eqn.

$$y(x) = C_1\, y_1(x) + C_2\, y_2(x)$$



Airy functions

In practice one usually works with the basis of *Airy functions*, defined by the linear combinations

$$\mathrm{Ai}(x) = \alpha\big[y_1(x) - \beta y_2(x)\big]$$

$$\mathrm{Bi}(x) = \sqrt{3}\alpha\big[y_1(x) + \beta y_2(x)\big]$$

where $\alpha \approx 0.355$ and $\beta \approx 0.729$.

Note $\mathrm{Ai}(x) \to 0$ as $x \to \infty$

We'll see $\mathrm{Ai}(x)$ again in Topic 6.

